

Time Series Diagnostic Report

Source: data/processed/AAPL_panel.parquet
Series: net_income_usd_m

Verdict

Forecastable:	Yes
Stationary:	No (difference d=1)
Seasonal:	Yes, period 4
Trend:	Increasing
Transform:	sqrt

Suggested starting model: SARIMA(p,1,q)(P,1,Q)[4] on sqrt(series)

Data quality

81 observations, quarterly, 2006-06-30 to 2026-06-30. Missing values: 0 (0.0%).

Mean 13523.42 Median 11253.00 Std 9849.87 Min 472.00 Max 42097.00

Skewness 0.6874 — noticeably skewed, which is why a transform is worth considering. Kurtosis -0.1163.

Stationarity

A model can only assume the future looks like the past if the series has a stable mean and variance. Two tests are run because they disagree in a useful way: ADF asks "is there a unit root?" and KPSS asks "is this stable around a level?". Agreement is strong evidence; disagreement usually means a trend rather than a random walk.

ADF statistic: **0.2846**

p = 0.9766. Above 0.05, so a unit root cannot be ruled out: the series wanders rather than reverting to a mean.

KPSS statistic: **1.3700**

p = 0.0100. At or below 0.05, so stability around a fixed level is rejected — consistent with a trend or drift.

Differencing needed: **d = 1**

This is the number of times to subtract the previous value before modelling. Differencing 1x removes the drift and is what the ACF/PACF plot is computed on. The result was re-tested and confirmed stationary.

Seasonality

Seasonality is a cycle that repeats at a fixed, known period. It is measured here by STL decomposition: the share of variation the repeating component explains, once the trend is removed. Strength runs 0 to 1.

Period: **4 (quarterly data)**

The cycle repeats every 4 observations, which for quarterly data is a full calendar year. Seasonal model terms are indexed on this number.

Seasonal strength: **0.8526**

strong — the cycle is a dominant feature and a seasonal model term is mandatory.

Trend

Trend is the slow movement of the level, separated from the repeating cycle by the same STL decomposition. A strong trend is the main reason a series is non-stationary and therefore why differencing is needed.

Direction: **increasing**

The level is increasing over the sample. Forecasts will extrapolate this, so satisfy yourself that the driver behind it still holds over the horizon — this is a judgement the statistics cannot make for you.

Trend strength: **0.9714**

Share of non-seasonal variation explained by the trend. Above 0.64 counts as strong: the trend dominates, and differencing is doing most of the work.

Normalized slope: **0.038926**

Average change per period, scaled by the series level, so it is comparable across series of different magnitudes.

Forecastability

Before fitting anything, it is worth knowing whether there is structure to fit. The Ljung-Box test asks whether the autocorrelations up to a given lag are jointly zero. A small p-value means they are not — past values carry information about future ones, so a model can beat a naive guess.

Ljung-Box (lag 10): **stat 153.29, p < 0.0001**

Structure detected — autocorrelation at these lags is not zero.

Ljung-Box (lag 16): **stat 256.07, p < 0.0001**

Structure detected — autocorrelation at these lags is not zero.

Ljung-Box (lag 20): **stat 300.31, p < 0.0001**

Structure detected — autocorrelation at these lags is not zero.

Verdict: **Forecastable**

The series is not white noise: a fitted model should outperform a naive forecast.

Autocorrelation (model order)

ACF and PACF are how the AR and MA orders get chosen. Read on the differenced series: the ACF tail suggests the MA order q , and the PACF cut-off suggests the AR order p . Only lags outside the confidence band count as real.

Series analysed: **differenced_d1**

Autocorrelation is only meaningful on a stationary series, so the plots use this version.

Confidence threshold: **0.2191**

Correlations smaller than this in absolute value are indistinguishable from zero at this sample size.

Significant ACF lags: **[1, 2, 3, 4, 5, 6, 8, 10, 12, 13]**

Candidate MA terms. Spikes at multiples of the seasonal period indicate a seasonal MA term rather than a long non-seasonal one.

Significant PACF lags: **[1, 2, 3, 4, 7, 8, 11, 13, 15, 20]**

Candidate AR terms. Start at the low end: the first few lags usually carry most of the signal, and extra terms cost degrees of freedom.

Variance transform

ARIMA assumes the size of the wobble does not depend on the level of the series. When swings grow as the series grows, a transform is applied first. Box-Cox estimates the exponent that best stabilizes the variance.

Variance stable: **No**

Spread grows with the level, so residuals from an untransformed model would be heteroscedastic and the prediction intervals too narrow at the high end.

Box-Cox lambda: **0.4690**

The optimal exponent is near 0.5, which corresponds to a square-root transform. Rounding to a standard transform is usually preferred over the exact value — it keeps the model interpretable and the back-transform simple.

Recommendation: **sqrt**

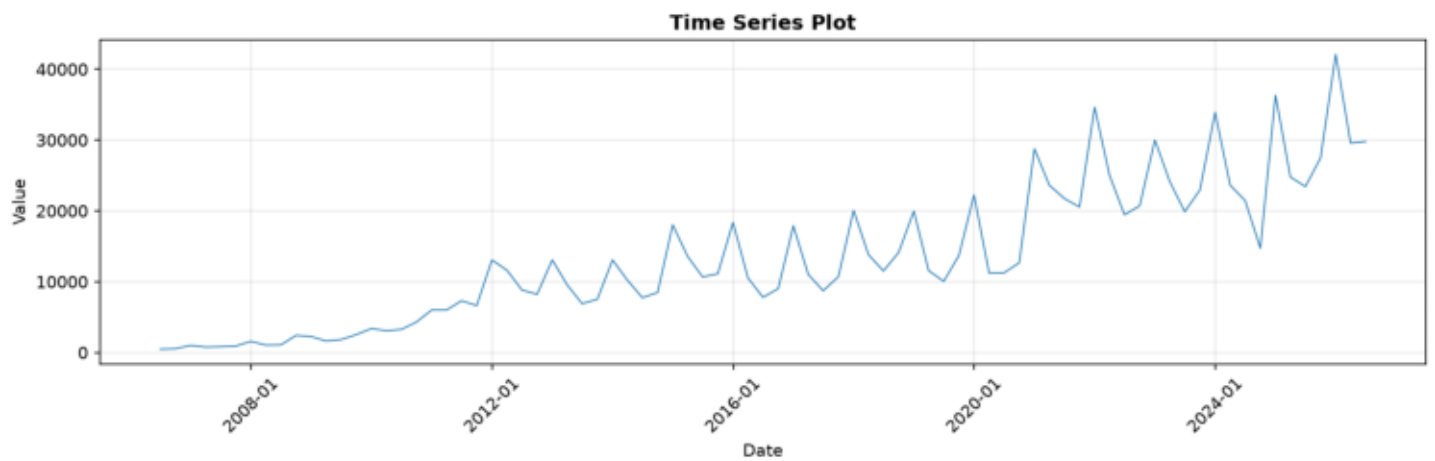
Apply this to the series before fitting, and remember to invert it on the forecasts and on both prediction-interval bounds.

Suggested next steps

1. Apply the sqrt transform first, and invert it on forecasts and interval bounds.
2. Start from SARIMA(p,1,q)(P,1,Q)[4] on sqrt(series) and refine p and q from the ACF/PACF plot.
3. Include seasonal differencing (D=1) at period 4: ordinary differencing alone will not clear correlation at the seasonal lags.
4. Hold out the most recent observations for validation rather than judging fit on the training sample — in-sample fit always flatters the model.
5. Check the residuals after fitting: they should pass Ljung-Box, meaning no structure was left on the table.

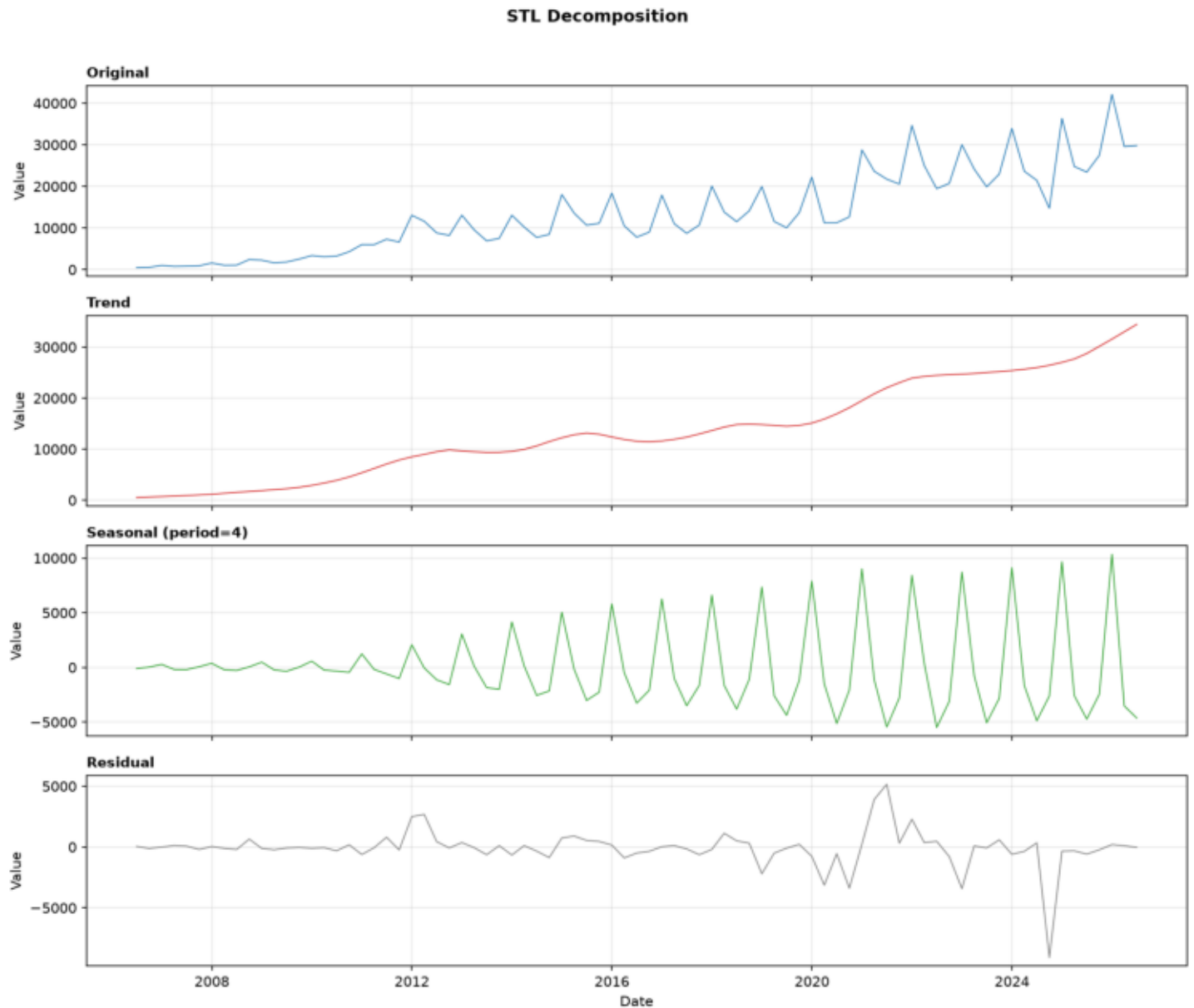
Timeseries

The raw series. Look for level shifts, outliers, and whether swings grow with the level.



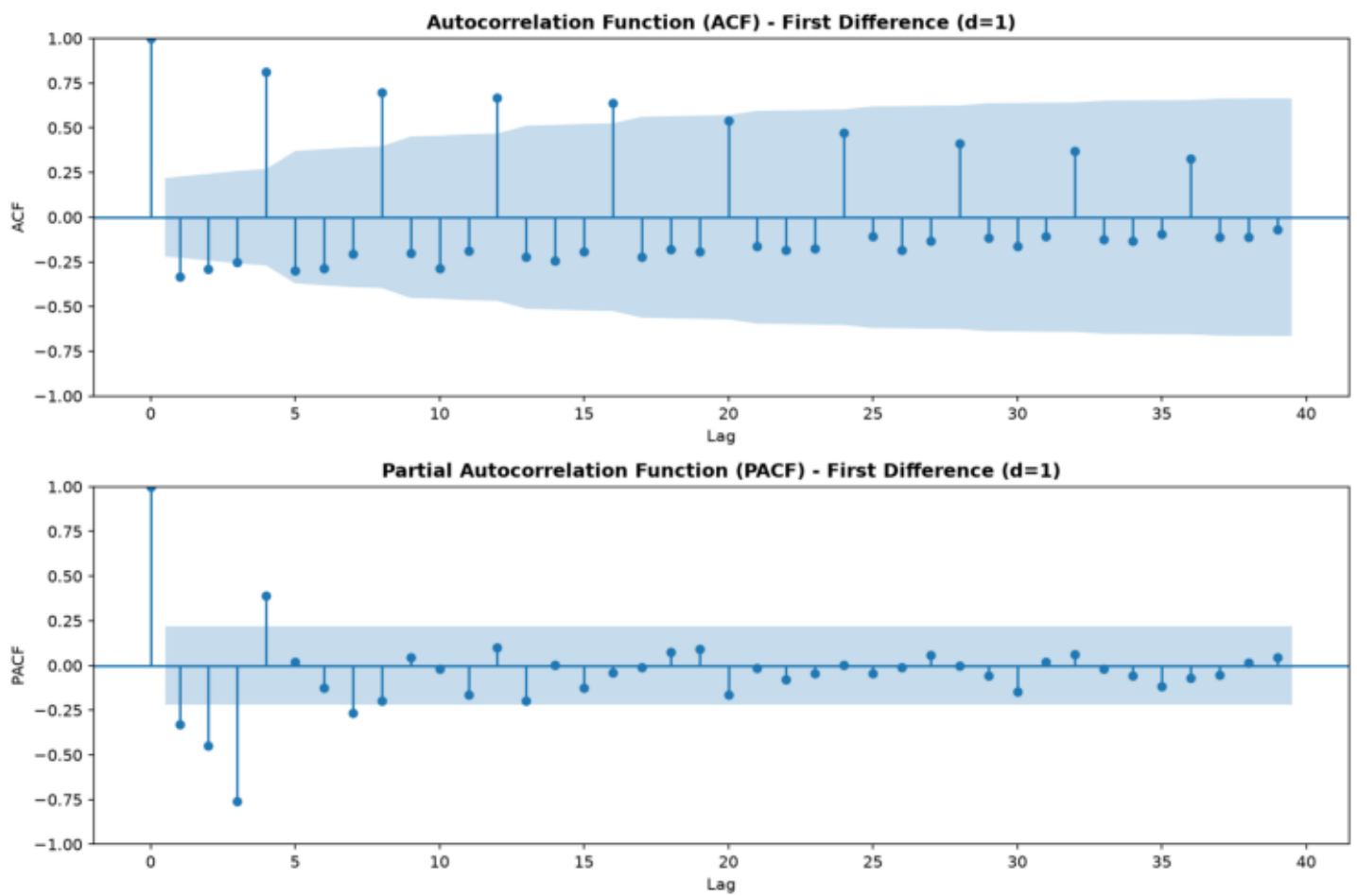
Decomposition

STL splits the series into trend, a repeating seasonal cycle, and the residual left over. A residual with visible structure means the split missed something.



Acf Pacf

ACF/PACF on the differenced series. Bars outside the shaded band are real correlation, and their positions suggest the AR and MA orders.



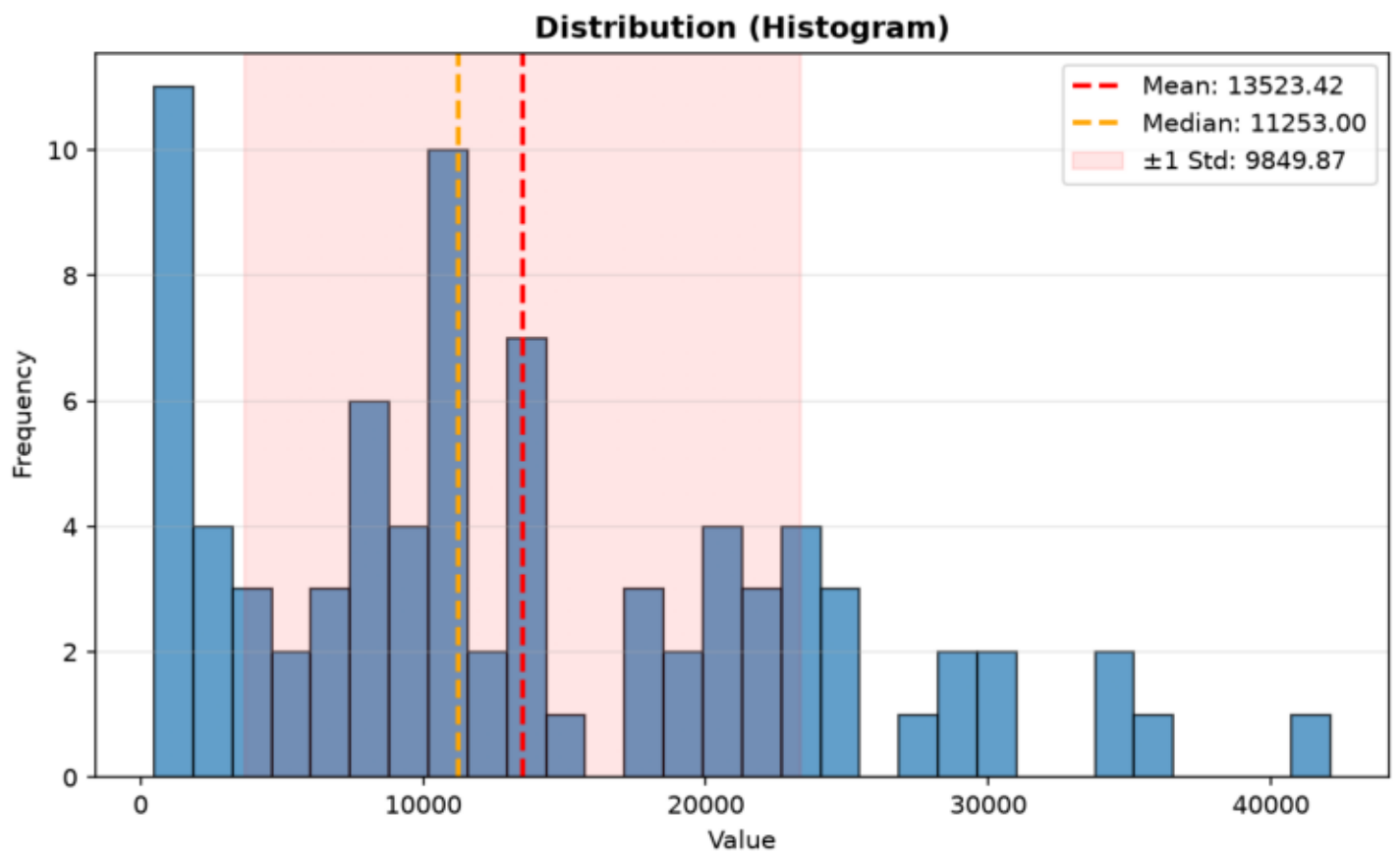
Rolling Stats

Rolling mean and standard deviation. A drifting mean means non-stationary; a fanning standard deviation is what a transform fixes.



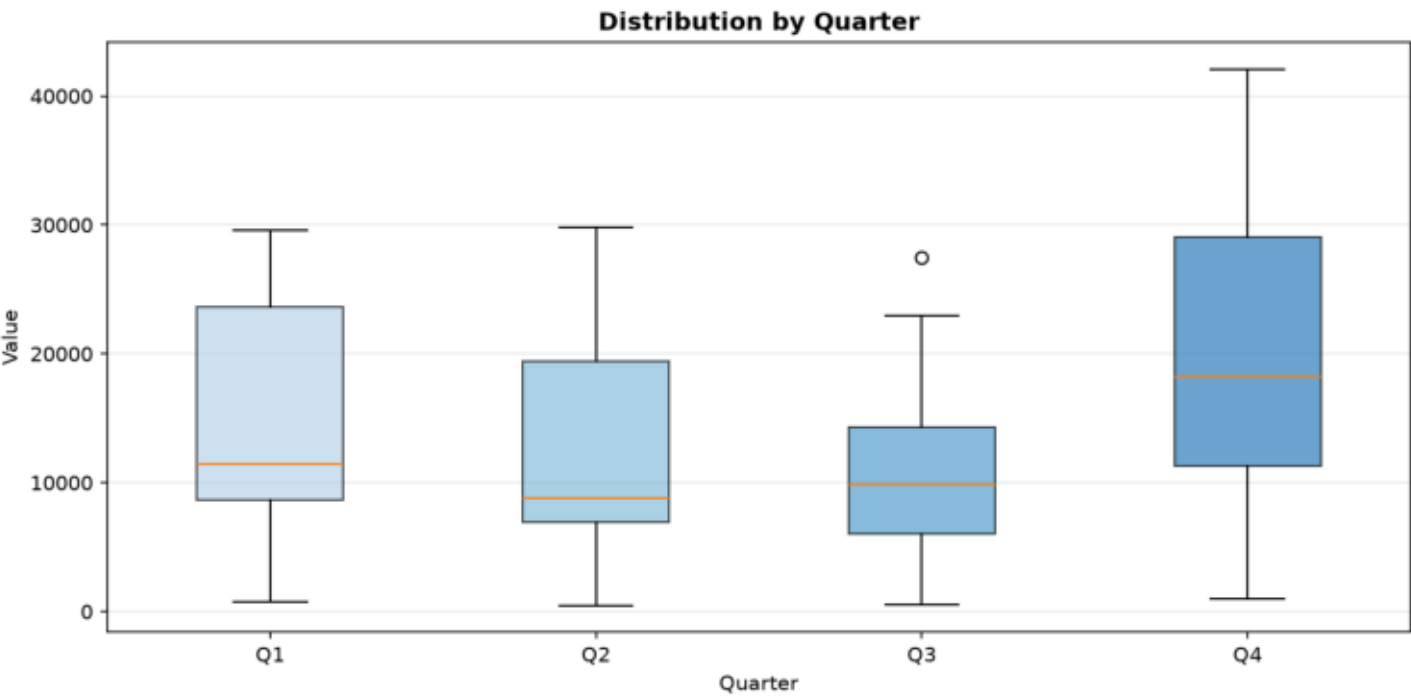
Histogram

Value distribution. Strong right skew is the usual sign that a log or sqrt transform will help.



Box By Quarter

Values grouped by quarter. Boxes sitting at clearly different heights confirm seasonality.



Lag Scatter

Each value against its own past. A tight diagonal means that lag carries real predictive signal.

